SATGURU STUDY CENTRE

Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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CLASS : 9

HALF YEARLY EXAMINATION-2024-25

MATHEMATICS

Register Number

Time Allowed : 3.00 Hours] [Max. Marks 100

PART - I

I Choose the correct Answer. 14x1=14

1 The set P = $\{x | x \in z, -1<x<1\}$ is a

a) Singleton set b) Power set c) Null set d) Subset

2 In a class of 50 boys, 35 boys play carrom and 20 boys play chess then the number of boys play both game is

a) 5 b) 30 c) 15 d) 10

3 For any three sets A, B and C, $(A - B) \cap (B - C)$ is equal to

a) A only b) B only c) C only d) ϕ

4 Which one of the following has a terminating decimal expansion?

a) $\frac{5}{64}$ b) $\frac{8}{9}$ c) $\frac{14}{15}$ d) $\frac{1}{12}$

5 If $\sqrt{80} = K\sqrt{5}$, then K =

a) 2 b) 4 c) 8 d) 16

6 The root of the polynomial equation $2x+3 = 0$ is

a) $\frac{1}{3}$ b) $\frac{-1}{3}$ c) $\frac{-3}{2}$ d) $\frac{-2}{3}$

7 Which of the following has x - 1 as a factor?

a) 2x - 1 b) 3x - 3 c) 4x - 3 d) 3x - 4

8 Cubic polynomial may have maximum of ------ linear factors

a) 1 b) 2 c) 3 d) 4

9 Degree of the constant polynomial is --------

a) 3 b) 2 c) 1 d) 0

10 The interior angle made by the side in a parallelogram is 90° then the parallelogram is a

a) Rhombus b) Rectangle c) Trapezium d) Kite

11 In a cyclic quadrilateral ABCD, $\angle A = 4x$, $\angle C = 2x$ then the value of x is

a) 30° b) 20° c) 15° d) 25°

12 The point whose ordinate is 4 and which lies on the y - axis is -----------

a) (4,0) b) (0,4) c) (1,4) d) (4,2)

13 The mid point of the line joining (-a, 2b) and (-3a, - 4b) is

a) (2a, 3b) b) (-2a, -b) c) (2a, b) d) (-2a, -3b)

14 The value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \ldots \tan 89^\circ$ is

a) 0 b) 1 c) 2 d) $\frac{\sqrt{3}}{2}$

PART - II

Answer any 10 questions. Question No. 28 is compulsory. 10x2=20

15 In n(A) = 4, Find n [P(A)]

16 Find the symmetric difference between the sets X = {5,6,7} and Y = {5, 7, 9, 10}

17 If n(A) = 300, $n(A\cup B) = 500$, $n(A\cap B) = 50$, and $n(B') = 350$, find n(B) and n(U)

18 Find the value of $(243)^{2/5}$

19 The mass of the Earth is 5.97×10^{24} Kg and that of the moon is 0.073×10^{24} Kg. What is their total mass?

CH9/A/Mat/1

20 Rewrite the polynomial $y^2 + \sqrt{5}\,y^3 - 11 - \frac{7}{3}y + 9y^4$ in standard form

21 If $P(x) = x^2 - 2\sqrt{2}x + 1$, find $P(2\sqrt{2})$

22 Factorise $(p - q)^2 - 6(p - q) - 16$

23 Find the GCD of $25ab^3c$, $100a^2bc$, $125ab$

24 The angles of a quadrilateral are in the ratio 2 : 4 : 5 : 7. Find all the angles

25 The centre of a circle is (-4, 2). If one end of the diameter of the circle's is (-3, 7) then find the other end

26 Find the centroid of the triangle whose vertices are A(6, -1), B(8, 3) and C(10, -5)

27 Evaluate $\sin 30^\circ + \cos 30^\circ$

28 A chord is 12 cm away from the centre of the circle of radius 15 cm. Find the length of the chord

PART - III

Answer the following any 10 questions. Q.No.42 is compulsory. **10x5=50**

29 Verify $A - (B \cup C) = (A - B) \cap (A - C)$ using venn diagrams

30 In a group of 100 students, 85 students speak Tamil, 40 students speak English, 20 students speak French, 32 speak Tamil and English, 13 speak English and French and 10 speak Tamil and French. If each student knows atleast anyone of these languages, Then find the number of students who speak all these three languages

31 Arrange in descending order $\sqrt[3]{5}, \sqrt[9]{4}, \sqrt[6]{3}$

32 Find the value of a and b if $\frac{\sqrt{7}-2}{\sqrt{7}+2} = a\sqrt{7} + b$

33 If both $(x - 2)$ and $(x - \frac{1}{2})$ are the factors of $ax^2 + 5x + b$, Then show that a = b

34 If $(x+a)(x+b)(x+c) = x^3 + 14x^2 + 59x + 70$, find the value of

i) $a+b+c$ ii) $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$ iii) $a^2 + b^2 + c^2$ iv) $\frac{a}{bc} + \frac{b}{ac} + \frac{c}{ab}$

35 Factorise $x^3 - 5x^2 - 2x + 24$

36 Solve by Cross multiplication method: $3x + 5y = 21$; $-7x - 6y = -49$.

37 Prove that in a parallelogram, opposite sides are equal

38 In a circle AB and CD are two parallel chords with centre O and radius 10 cm such that AB = 16 cm and CD = 12 cm, Determine the distance between the two chords

39 Show that the points A (3, 1), B (6, 4), and C (8, 6) lies on a straight line

40 The mid point of the sides of the triangle are (2,4), (-2, 3) and (5,2). Find the co ordinates of the vertices of the triangle.

41 Find the co - ordinates of the points of trisection of the line segment joining the points A (-5,6) and B (4,-3)

42 If $2\cos\theta = \sqrt{3}$, then find all the trigonometric ratios of angle θ.

PART - IV

Answer all the questions. **2x8=16**

43 a) Construct the centroid of ΔPQR whose sides are PQ = 8cm, QR = 6cm, RP = 7cm.

(OR)

b) Draw ΔPQR with sides PQ = 7cm, QR = 8cm and PR = 5cm and Construct its Orthocentre

44 a) Draw the graph of $y = 3x - 1$

(OR)

b) Solve graphically: $x + y = 7$; $x - y = 3$